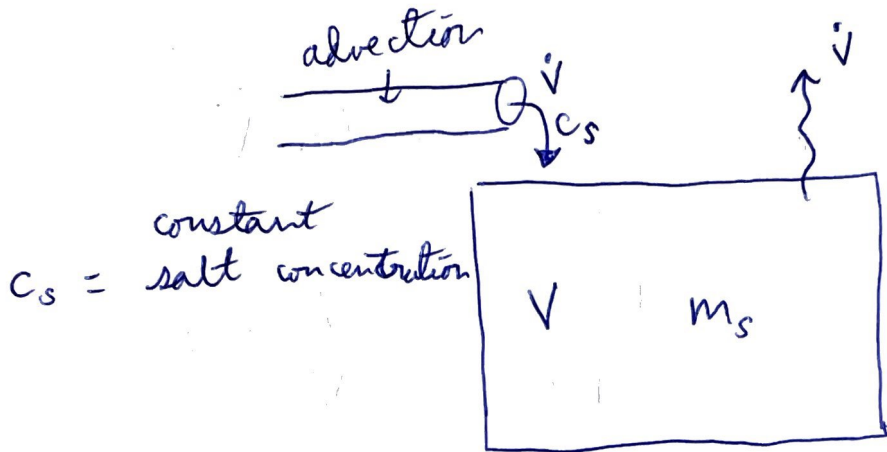


| Salinity solution |

(S1)

Conceptual model

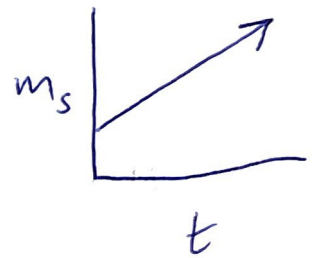


$\dot{V} \equiv$
Constant runoff
(river flow) rate =
evaporation rate

~~no~~ No loss of
salt!

Fixed volume
(constant)

$$\dot{m}_s \equiv \text{salt flow in} = \dot{V} \cdot c_s$$



Conservation equation

salt accum. = flow in

Constitutive equation

$$\dot{m} = \dot{V} \cdot c_s$$

GE (governing equation)

(32)

accum. = flow in

$$\frac{d m_s}{d t} = \dot{V} C_s$$

or divide by V

$$\frac{d C_s}{d t} = \frac{\dot{V}}{V} C_s$$

Solution

Integrate GE:

constant of integration, here =
init salt conc.

$$C_s(t) = C_{s,0} + \left(\frac{\dot{V}}{V} \right) \cdot C_s \cdot t$$

Turnover rate or

Application

$\frac{1}{\tau} \rightarrow$ retention time

$$C_{s,0} \approx 35 \text{ ppt} = 3.5\% \approx 35 \text{ g/L}$$

$$\frac{\dot{V}}{V} = \frac{50 \cdot 10^3 \text{ km/yr}}{1.4 \cdot 10^9 \text{ km}} = 3.6 \cdot 10^{-5} \text{ yr}^{-1}$$

$$C_s(10^3) = 35 \text{ g/L} + \underbrace{0.01 \text{ g/L} \cdot 10^3 \text{ yr}}_{0.00036 \text{ g/L}}$$